

15-2c Profit Maximization

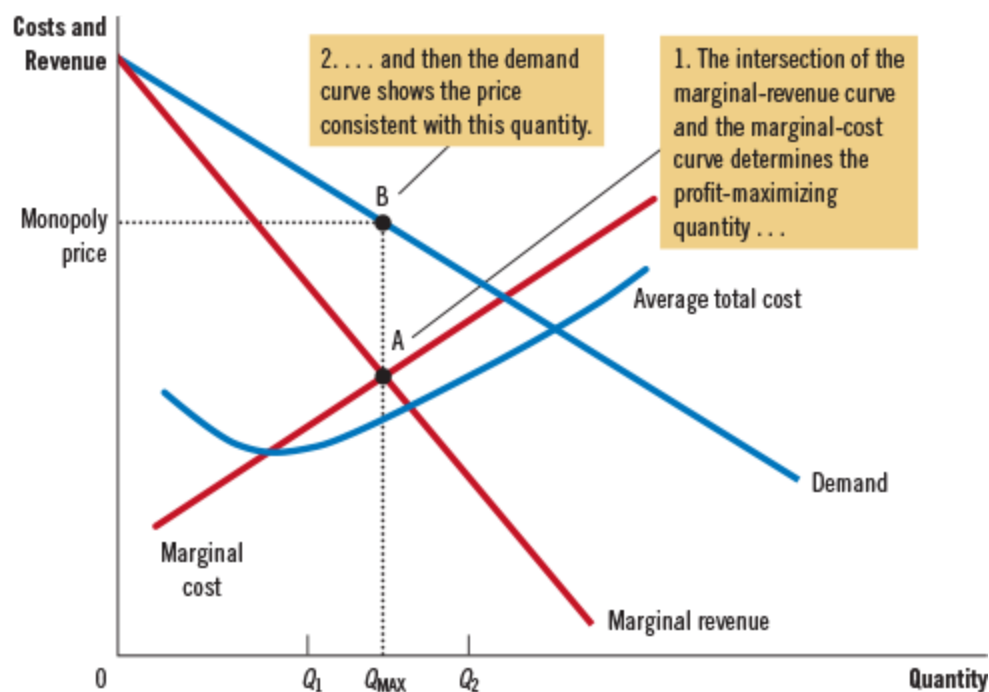
Now that we have considered the revenue of a monopoly firm, we are ready to examine how such a firm maximizes profit. Recall from [Chapter 1](#) that one of the *Ten Principles of Economics* is that rational people think at the margin. This lesson is as true for monopolists as it is for competitive firms. Here we apply the logic of marginal analysis to the monopolist's decision about how much to produce.

[Figure 4](#) graphs the demand curve, the marginal-revenue curve, and the cost curves for a monopoly firm. All these curves should seem familiar: The demand and marginal-revenue curves are like those in [Figure 3](#), and the cost curves are like those we encountered in the last two chapters. These curves contain all the information we need to determine the level of output that a profit-maximizing monopolist will choose.

Figure 4

Profit Maximization for a Monopoly

A monopoly maximizes profit by choosing the quantity at which marginal revenue equals marginal cost (point A). It then uses the demand curve to find the price that will induce consumers to buy that quantity (point B).



Suppose, first, that the firm is producing at a low level of output, such as Q_1 . In this case, marginal cost is less than marginal revenue. If the firm were to increase production by 1 unit, the additional revenue would exceed the additional costs, and profit would rise. Thus,

when marginal cost is less than marginal revenue, the firm can increase profit by producing more units.

Similar reasoning applies at high levels of output, such as Q_2 . In this case, marginal cost is greater than marginal revenue. If the firm were to reduce production by 1 unit, the costs saved would exceed the revenue lost. Thus, when marginal cost is greater than marginal revenue, the firm can raise profit by producing fewer units.

In the end, the firm adjusts its level of production until the quantity reaches Q_{MAX} , at which marginal revenue equals marginal cost. Thus, *the monopolist's profit-maximizing quantity of output is determined by the intersection of the marginal-revenue curve and the marginal-cost curve*. In Figure 4, this intersection occurs at point A.

You might recall from the previous chapter that competitive firms also choose the quantity of output at which marginal revenue equals marginal cost. In following this rule for profit maximization, competitive firms and monopolies are alike. Yet there is an important difference between these types of firms: The marginal revenue of a competitive firm equals its price, whereas the marginal revenue of a monopoly is less than its price. That is,

For a competitive firm: $P = MR = MC$.

For a monopoly firm: $P > MR = MC$.

The equality of marginal revenue and marginal cost determines the profit-maximizing quantity for both types of firm. But unlike firms in a competitive market, a monopoly firm charges a price above marginal revenue and marginal cost.

So how does the monopoly find the profit-maximizing price for its product? The demand curve gives the answer because it relates the amount that customers are willing to pay to the quantity sold. After the monopoly firm finds the profit-maximizing quantity (at which $MR = MC$), it looks to the demand curve to find the highest price it can charge at that quantity. In Figure 4, the profit-maximizing price is found at point B.

We can now see a key difference between markets with competitive firms and markets with a monopoly firm: *In competitive markets, price equals marginal cost. In monopolized markets, price exceeds marginal cost*. As we will see in a moment, this finding is crucial to understanding the social cost of monopoly.